

ITSM Copilot and Support Automation

A practical service-desk workflow for intake, triage, case creation, approvals, escalation, and reporting using AI Builder, Azure AI Foundry, Azure Functions, Power Automate, and Power BI.

Sanitized implementation case study

01

Support intake hub

L2/L3

Escalation path

SLA

Priority-aware routing

Power BI

Operational reporting

What this shows: Support requests arrived as free-text messages and manual forms. Agents had to interpret the issue, ask for missing information, decide priority, create a case, and hand it to the correct queue. This created inconsistent categorization and made it difficult to track response and resolution performance.

End-to-end workflow

1

Employee request

Source enters the process.

2

Copilot topic

Context and missing fields are identified.

3

Intent + priority

Data becomes consistent and usable.

4

Case + queue

Rules compare, route, or reconcile.

5

Resolution / handoff

People receive an action-ready result.

Business value

- Transforms unstructured or incomplete source information into consistent records.
- Separates automation from human review so exceptions are visible.
- Creates a traceable path from original source to report or action.
- Supports operational teams with clearer ownership and faster review.

How to read this project

HR can review the first two pages for the business process and my contribution. The linked repository folder contains sanitized data models, workflow examples, documentation, and implementation notes for deeper technical review.

How the solution works

The important part of this project is the complete operating chain: source capture, structured transformation, business-rule processing, exception handling, and reporting. Each stage produces a clear output that can be validated before the next stage uses it.

01 / INTAKE	The Copilot Studio agent recognizes common support topics such as access requests, incidents, knowledge lookup, case status, and request-for-agent. It asks guided follow-up questions for requester, system, urgency, and business impact. Output: traceable, reviewable data for the next stage
02 / TRIAGE	A structured routing payload carries intent, extracted entities, priority, confidence, and required next action. High-confidence requests continue automatically; low-confidence or sensitive requests are flagged for human review. Output: traceable, reviewable data for the next stage
03 / ORCHESTRATION	Power Automate validates required fields, applies routing and approval rules, creates or updates the Dataverse-style case record, generates an SLA target, and sends the assigned queue an Adaptive Card. Output: traceable, reviewable data for the next stage
04 / SUPPORT HANDOFF	If automation cannot resolve the request, the handoff packet includes the conversation summary, attempted steps, extracted entities, attachments, confidence score, and reason for escalation so L2/L3 support can continue without repeating discovery. Output: traceable, reviewable data for the next stage
05 / REPORTING	Resolution status, response time, topic performance, escalation reason, and feedback are captured for Power BI or Microsoft Fabric-style reporting and continuous improvement. Output: traceable, reviewable data for the next stage

Example structured record

```
{ "source_reference": "BATCH-2026-001 / DOCUMENT-0042", "record_id": "REC-0042", "status": "review_required", "captured_fields": { "account_number": "sample-account", "document_number": "sample-document", "amount": 0, "tax": 0 }, "checks": { "required_fields": "passed", "duplicate": "not_found", "reconciliation": "variance_found" }, "next_action": "finance_review" }
```

Design principle: Never hide an exception inside a final total. Keep the source reference, extracted value, expected value, variance, and review status together so an operations user can explain what happened.

Data and control points

Control point	Purpose	Typical output
Source reference	Connects the structured row to the physical or digital document.	Batch ID, file ID, page or document reference
Validation	Prevents missing, malformed, duplicate, or low-confidence data from flowing silently.	Pass, fail, review required
Business rules	Applies finance or service logic to compare, route, calculate, or approve.	Matched, variance, assigned queue
Exception record	Makes human review explicit and measurable.	Expected, actual, variance, owner, status
Reporting layer	Turns operational records into trends and actions.	Volume, aging, variance, SLA, resolution

Responsibilities and technology map

My contribution

- Mapped the operational process from source document or user request to final report.
- Defined fields, validation checks, exception states, and review ownership.
- Designed automation steps and structured payloads for repeatable processing.
- Connected data preparation with reporting and operational follow-up.
- Documented the workflow so another team member can understand, test, and operate it.

Portfolio boundaries

This is a sanitized portfolio representation. Casino/client names, real account numbers, production URLs, credentials, and confidential records are intentionally excluded. The domain workflow and technical responsibilities are preserved at a useful level for discussion.

Capability matrix

Capability	Evidence in this project	Keywords
Process automation	Multi-stage intake, transformation, validation, routing, exception, and reporting flow.	Power Automate, workflow design, integration
Data engineering	Structured fields, normalization, staging tables, checks, and downstream reporting data.	Python, SQL, Excel, Power Query, Dataverse
Finance operations	Document capture, transaction fields, reconciliation, variance identification, review states.	OCR, reconciliation, controls, audit trail
Conversational / AI patterns	Where applicable, intent, confidence, fallback, and human-review concepts.	Copilot Studio, Azure AI, Azure OpenAI
Reporting	Operational metrics, variance categories, aging, SLA, resolution and trend views.	Power BI, Microsoft Fabric, dashboards

Technology stack

Copilot Studio

Power Automate

Power Apps

Dataverse

AI Builder

Azure AI Foundry

Azure OpenAI

Azure Functions

Adaptive Cards

Power BI

Microsoft Fabric

App Insights

REST APIs

JSON

C# / .NET

JavaScript / TypeScript

Git / CI/CD

Agile / Scrum

Suggested interview walkthrough: Start with the source document or request, explain how you preserved traceability, show the validation and reconciliation decision points, then explain how the final report helped an operations or support user take action.